IoT Smart Home — Voice-Controlled Lighting

Public-safe technical overview | MIT109 academic prototype

Project overview

This project demonstrates voice-controlled lighting using a browser speech interface, a Node-RED HTTP-to-MQTT gateway, a TLS-capable MQTT broker, and an ESP32 microcontroller simulated in Wokwi. The ESP32 switches an LED connected to GPIO 23 when it receives an accepted ON or OFF command.

Architecture

- Browser speech recognition maps supported phrases to ON/OFF commands.
- A separately configured Node-RED gateway validates incoming requests and publishes accepted commands.
- An MQTT broker forwards messages to the subscribed ESP32 client.
- The ESP32 controls the simulated LED; other commands are rejected.

The editable Draw.io diagram and PNG preview are supplied alongside this document in the docs folder.

Security and implementation notes

The original submitted firmware contained embedded MQTT credentials and disabled server-certificate verification. The public GitHub version instead uses a private config.h file and a trusted CA certificate template. Previously exposed MQTT credentials must be rotated before reuse.

The Node-RED gateway is a required external component. Its complete importable flow was not included in the source archive, and authentication, CORS restrictions and HTTPS must be configured before internet exposure.

Verification and limitations

The original coursework report describes Wokwi, MQTT and Node-RED demonstrations. Those historical screenshots are deliberately omitted from this public copy because they may disclose broker or environment details. GitHub Actions provides static source checks only; it does not prove that the broker, gateway and Wokwi simulation currently operate together.

A complete end-to-end verification requires fresh MQTT credentials, a running gateway, an accessible broker and a compatible browser with microphone permission.

References

- Banks, A. and Gupta, R. (2014). MQTT Version 3.1.1. OASIS. https://docs.oasis-open.org/mqtt/mqtt/v3.1.1/
- Rescorla, E. (2018). The Transport Layer Security (TLS) Protocol Version 1.3. RFC 8446. https://datatracker.ietf.org/doc/html/rfc8446
- MDN Web Docs. Web Speech API. https://developer.mozilla.org/en-US/docs/Web/API/Web_Speech_API